

2-4 page revision guide: terminology, methods, comparisons and common exam mistakes

Revision aid

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All datasets used in this package are simulated teaching data. Wet-lab and animal procedures described here are conceptual; institutional biosafety approval and SOPs govern all practice.

Cheat Sheet — GMOs, Cloning & Gene Expression

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1. Essential terminology

Term	One-line definition
Genetic engineering	Deliberate, direct modification of genetic material using lab techniques
Recombinant DNA	DNA joined from >1 source into a propagatable molecule
Molecular cloning	Building + amplifying a defined DNA molecule in a host
GMO	Organism with lab-altered genetic material (definition is regulatory!)
Transgenic	Carries foreign (cross-species) DNA
Cisgenic	Carries same-species DNA
Gene-edited	Targeted editor (CRISPR etc.) made the change
Knockout / Knock-in	Gene inactivated / defined sequence inserted at a locus
Stable vs transient	Inherited vs fading (plasmid/integration vs transfection)
Expression cassette	promoter → 5' UTR/RBS → CDS → terminator/polyA

2. Cloning vocabulary

Insert (passenger) · **Vector** (vehicle) · **Backbone** (vector minus insert) · **ori** (replication origin; sets copy number) · **MCS** (multi-cloning site) · **Selectable marker** (cell survives only with vector) · **Screenable marker** (distinguishes clones visually) · **Fusion protein** (in-frame tag+protein) · **Shuttle vector** (two hosts) · **BAC/YAC** (100–300 kb / ~1 Mb capacity).

3. Vector components — what breaks without them

Component	Missing →
ori	no propagation
Selectable marker	can't select transformants
MCS	insertion impractical
Promoter/RBS/Kozak	clone but no expression
Terminator/polyA	read-through / unstable transcripts
Reporter	expression invisible

4. Restriction enzymes & ligase

- Recognition sites 4–8 bp, usually palindromic; **EcoRI** G↓AATTC (sticky 5'-AATT), **EcoRV** blunt, **BamHI** G↓GATCC, **HindIII** A↓AGCTT.
- **Sticky vs blunt ends**; **compatible ends** (BamHI/BglII → same overhang).
- **T4 DNA ligase** seals 5'P–3'OH nicks (ATP-dependent).
- **Dephosphorylation** of vector blocks self-ligation.
- **Insert:vector ratio** ~3:1 (molar) for sticky-end ligations (start point, not law).
- Methylation sensitivity can block some sites in lab *E. coli* DNA.

5. Cloning methods at a glance

Method	Principle	Best for
Restriction-ligation	cut+anneal+ligate	single insert, directional
TA	Taq A-overhang ↔ T-vector	fast PCR capture
Blunt	flush-end ligation	no sites available
Gibson	exonuclease chew-back + overlap anneal + fill + seal	seamless multi-fragment
Golden Gate	Type IIS → designer 4-bp overhangs, one-pot	modular, multiplex
Gateway	att-site recombination (entry → destination)	one entry → many vectors

Golden rule: Gibson/Golden Gate = design-driven addressing; restriction = site-driven; Gateway = recombination-driven.

6. Selection vs screening vs validation

Selection → only vector-bearing cells grow (antibiotic/auxotrophy)

Screening → which colonies have the RIGHT vector (blue-white, colony PCR)

Validation → prove structure (digest) then sequence (Sanger), then function

- **Blue-white:** white = insert-disrupted lacZ α (candidate, not proof).
- **Colony PCR:** vector+insert primer combo = orientation readout.
- **Diagnostic digest:** fragments must match map AND sum to plasmid size.
- **Sanger:** junctions + full insert coverage; final word.

7. Reporter genes

Reporter	Signal	Needs substrate?	Best at
GFP	fluorescence ~508 nm	no (needs O $\blacksquare$)	live spatial imaging
mCherry/RFP	red fluorescence	no	dual-color with GFP
lacZ	blue (X-gal)	yes	fixed-tissue histology
Luciferase (firefly)	photons	luciferin+ATP	quantitative dynamics

Controls: promoterless & empty vector; positive promoter; motif mutant; normalization (dual-luciferase / co-transfection). **Caveat:** reporter = regulatory-sequence behavior, not necessarily endogenous gene behavior.

8. GMO terminology recap

Transgenic ≠ cisgenic ≠ gene-edited. **The reagent is transient; the edit can be stable.** SCNT (Dolly) = cloning, no new DNA. Random integration (classic transgenesis) vs targeted edit (CRISPR era).

9. CRISPR terminology

- **sgRNA** ~20-nt guide + scaffold; **PAM** (SpCas9: NGG) required for recognition.
- **DSB** → **repair:** **NHEJ** = indels = knockout; **HDR** = donor template = knock-in/correction.
- **Base editors** (C→T, A→G) and **prime editors** = no DSB.
- **dCas9** = binding-only chassis → **CRISPRa** (activate) / **CRISPRi** (repress, e.g., KRAB).
- Pitfalls: off-targets, mosaicism (F0), in-frame indels escaping knockout.

10. Developmental gene regulation

- Differential gene expression = same genome, different transcriptional programs.

- **Cell fate** → **determination** → **differentiation**; potency: totipotent → pluripotent → multipotent.
- **Morphogen gradient + thresholds** (French-flag): Shh (neural tube DV, limb ZPA), Bicoid (Drosophila A–P), BMP (dorsal ectoderm), Wnt (axis), RA (hindbrain/limb PD).
- **GRN motifs**: positive feedback (lock fate), mutual repression (sharp borders), feed-forward (filter), lateral inhibition (salt-and-pepper, Notch).
- Landmark genes: **Hox** (colinear A–P identity), **Pax6** (eye field), **Shh**, **Wnt1**, **myoD** (myogenic master — sufficiency caveat).

11. Spatial vs temporal expression

Question	Method of choice
Where (endogenous mRNA)?	ISH / RNA-FISH
Where (protein)?	IHC / IF
Where + dynamics?	Reporter (GFP/luc)
How much over time (few genes)?	RT-qPCR ($\Delta\Delta Ct$)
Which genes define a cell type?	scRNA-seq
Which genes define a region?	spatial transcriptomics

Gradient = continuous concentration field; **domain** = bounded region. Gradients inform; domains execute.

12. Experimental-method quick table

Method	Level	Resolution	Quantitative?
RT-qPCR	mRNA	none	yes (relative)
Bulk RNA-seq	transcriptome	none (mean)	relative
scRNA-seq	transcriptome	cell (position lost)	relative, sparse
Spatial transcriptomics	transcriptome	spot/cell	relative
smFISH	mRNA	subcellular	absolute counts
ISH	mRNA	cell/tissue	semi
IHC/IF	protein	subcellular	semi
Reporter	regulatory output	cell/tissue, live	semi-good

13. Data-analysis concepts

- $\Delta\Delta Ct$: $\Delta Ct = Cq(\text{target}) - Cq(\text{ref})$; $FC = 2^{-(\Delta\Delta Ct)}$. Stats on ΔCt , not FC.
- **RNA-seq**: normalize (CPM/size factors) → model → **FDR** (BH). $|\log_2 FC|$ with FDR filter; beware low counts.
- **scRNA**: QC (library, genes, doublets) → normalize/log → PCA → cluster → **marker annotation**. Dropout ≠ absence.
- **Spatial**: matrix + coordinates → region markers → gradient test (correlation with distance).
- **Heatmaps**: z-score genes to compare *patterns* regardless of magnitude.
- Report: n, normalization, test, effect size, version. Always.

14. Important comparisons (rapid-fire)

- Transgenesis **adds**; editing **changes**.
- Selection **enriches populations**; screening **classifies individuals**; validation **proves identity**.
- Gradient **informs**; domain **executes**.

- Fluorescence **spatial**; luminescence **quantitative**.
- NHEJ **breaks**; HDR **writes**.
- Bulk **averages**; single-cell **resolves types**; spatial **resolves places**.
- CRISPRa tests **sufficiency**; CRISPRi/knockout tests **necessity**.

15. Common exam mistakes

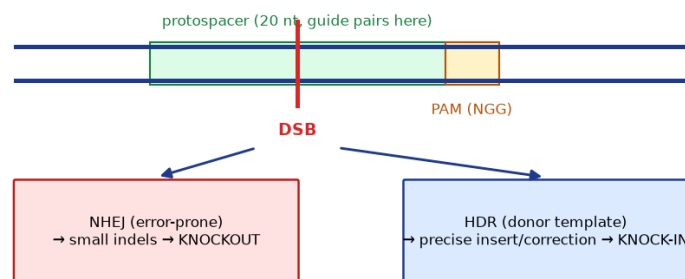
1. Calling any transgenic organism a "clone" (or vice versa — Dolly wasn't transgenic).
2. Reading colony count as correct-clone count (screening is separate!).
3. Blue colony = always empty / white = always correct (exceptions exist).
4. Confusing PAM (genomic side) with protospacer (guide-target region).
5. Saying CRISPR "inserts a gene" by default (NHEJ knocks out; knock-ins need HDR/donors or editors).
6. $\Delta\Delta C_t$ applied to unstable reference genes.
7. Interpreting a reporter domain as the endogenous pattern without ISH validation.
8. Confusing gradient (continuous) with domain (bounded).
9. Treating dropout in scRNA-seq as biological absence.
10. Claiming an enhancer is "necessary" from a reporter (sufficiency only).
11. Summing digest fragments is skipped → rearranged plasmids pass unnoticed.
12. Confusing cis-regulatory elements (on the DNA) with trans-acting factors (the proteins that bind them).

Print tip: this file paginates to ~4 pages at A4 in the generated PDF. Related:

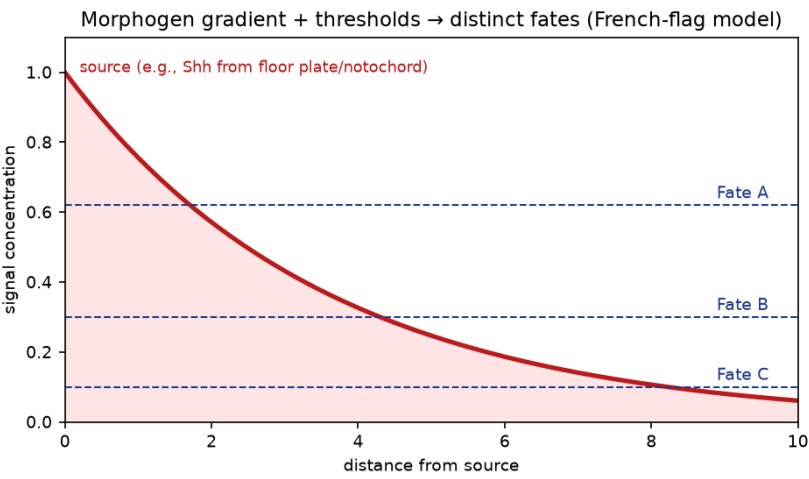
[Workbook](../WORKBOOK/GMO-Cloning-Gene-Expression-Lab-Workbook.md) ·

[References](../REFERENCES.md)

CRISPR-Cas9: target recognition and repair outcomes



CRISPR-Cas9 target recognition and repair outcomes (revision figure).



Morphogen gradient with threshold fates (revision figure).